ICT2113 · Data Structures and Algorithms

Worked Examples: Search and Sorting

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Linear search checks each element and needs at most n comparisons. Binary search repeatedly halves a sorted search range, giving logarithmic comparisons, but only when the sorted-order precondition holds.

2. Example and reasoning

For a sorted sequence of 32 values, binary search needs at most about five halvings to isolate a value. Merge sort divides, recursively sorts, and merges runs; its predictable O(n log n) time is useful when stable ordering matters.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “worked examples: search and sorting” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.